

Finite Element Modeling & Geometric Optimization for Ultrasonic Fatigue

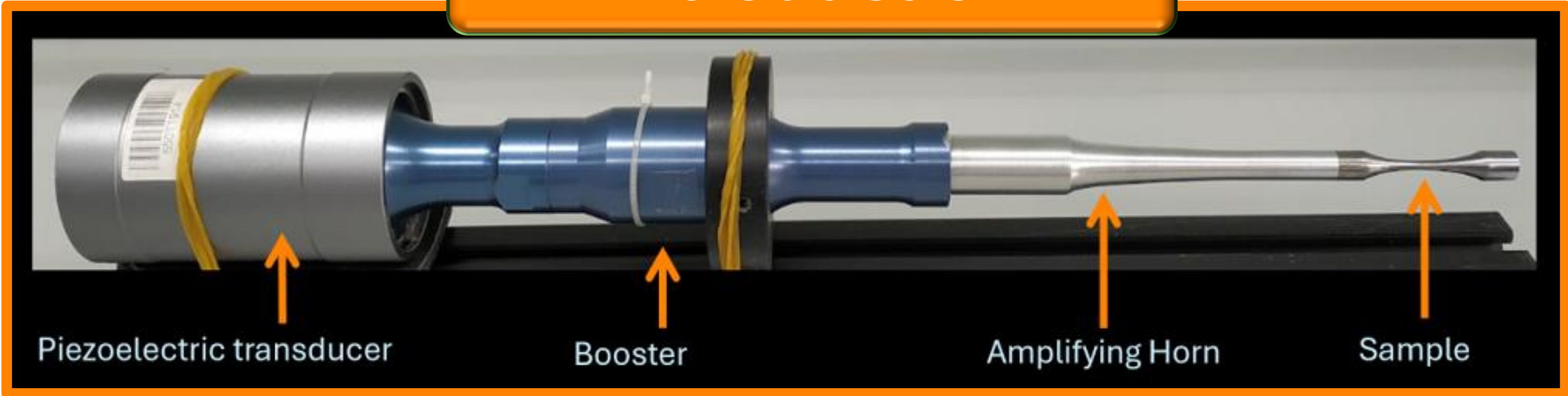
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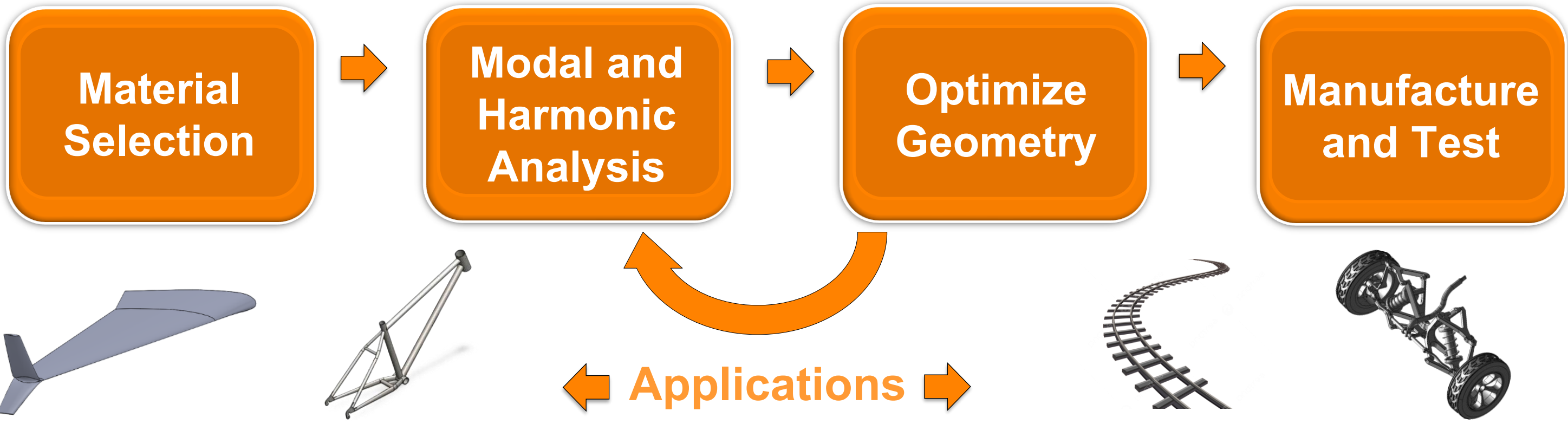
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Introduction



Modeling & Optimization Workflow



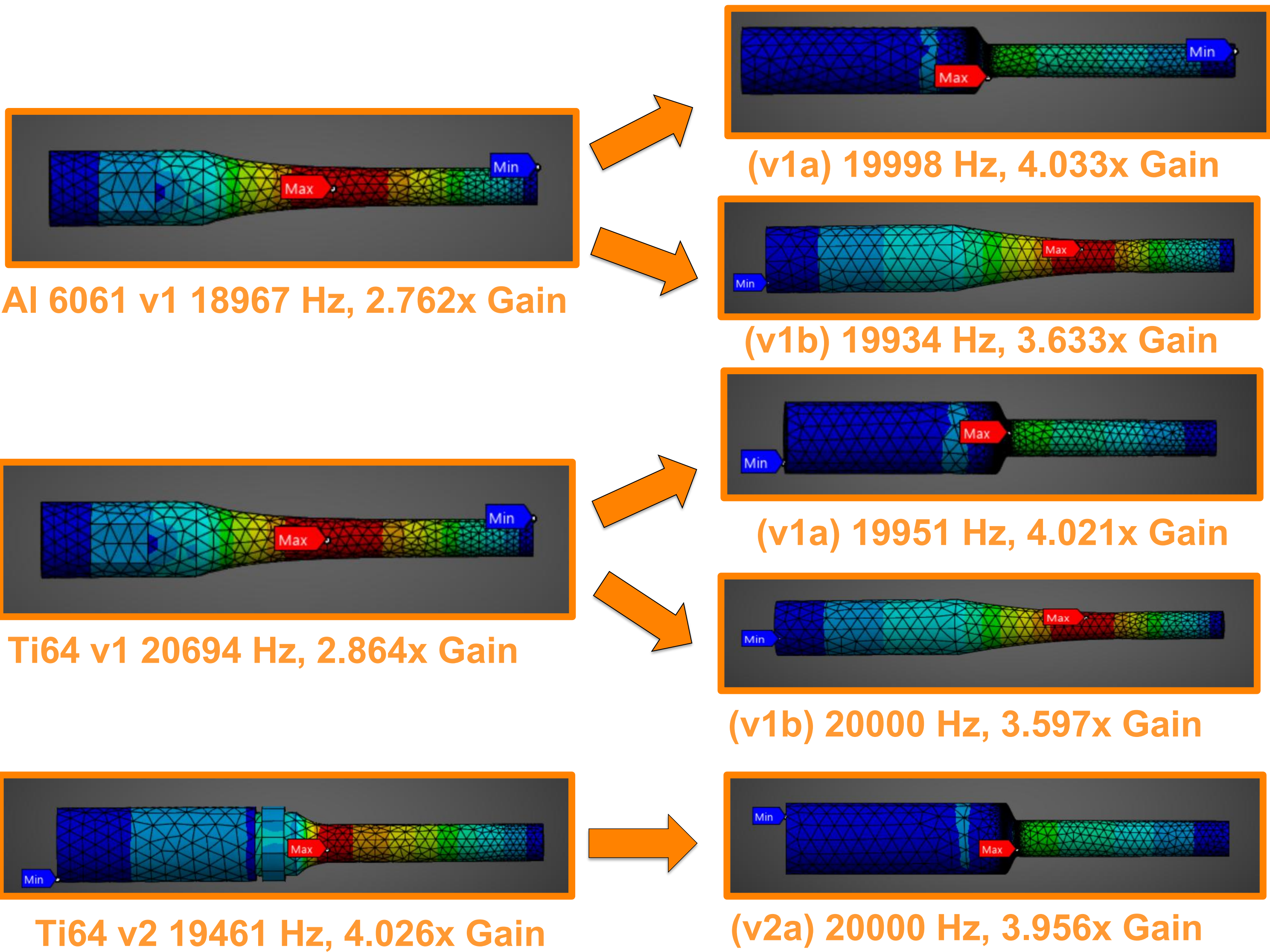
Objective

Why did some work and why did some fail?

How to make the most optimal design?

What is the most optimal design?

Amplifying Horn Design



Design Theory

Wave Speed:

$$c = \sqrt{\frac{E}{\rho}}$$

Half-Wave Length:

$$\frac{\lambda}{2} = \frac{c}{2f}$$

Nodal Stress:

$$\sigma_{node} \approx \rho c \cdot \omega \cdot u_{tip}$$

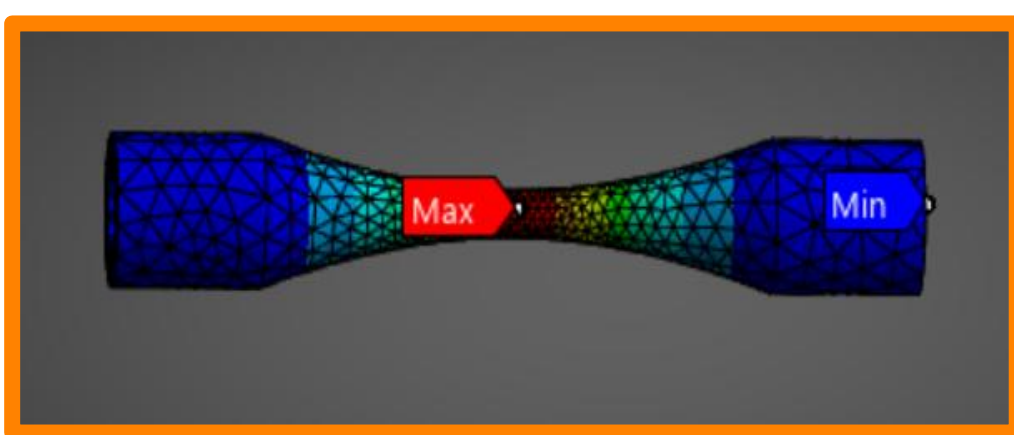
Gain Ceiling:

$$M_{max} = \left(\frac{D_{base}}{D_{tip}} \right)^2$$

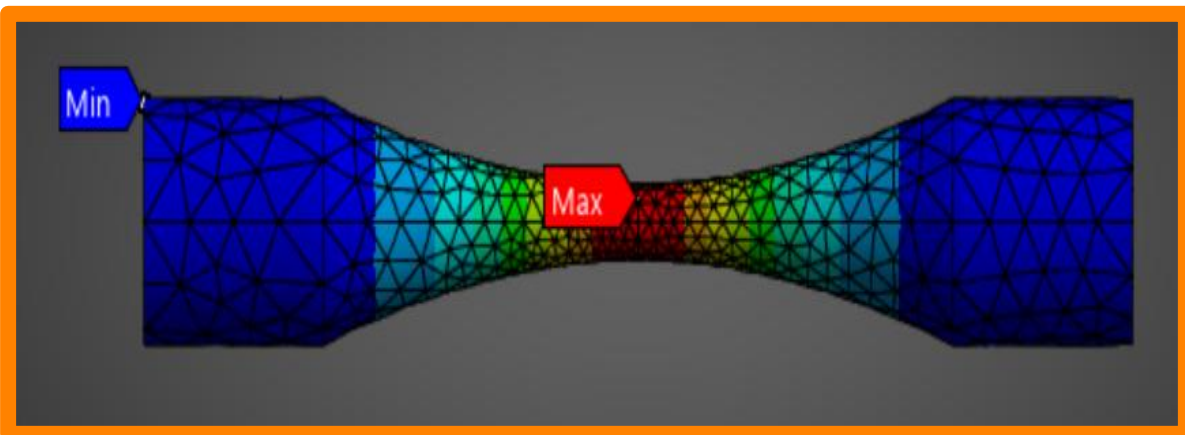
Gain from Design:

$$M = \left| \frac{u_{tip}}{u_{base}} \right|$$

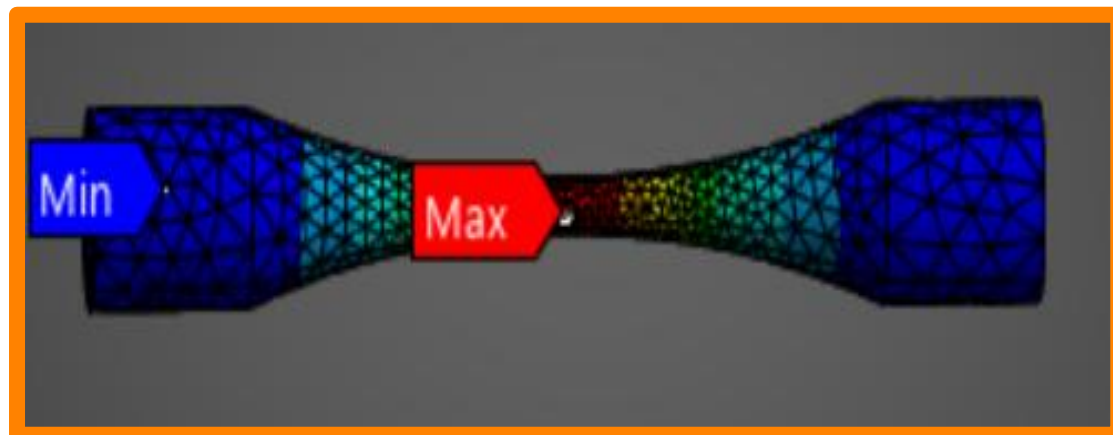
Hourglass Design



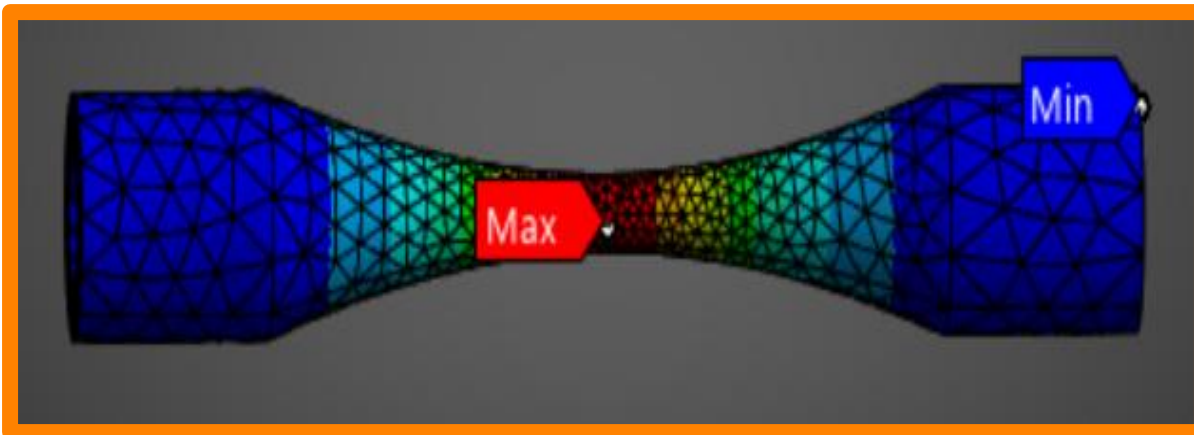
4140 Carbon Steel
19959 Hz



Stainless Steel 316
19957 Hz



Al 6063-T52
19976 Hz



Al 2024-T4
19983 Hz

Hourglass Dimensions Before & After

Hourglass	Original Length	Optimized Length	Original Resonance	Optimized Resonance
4140 Carbon Steel	61.60 mm	61.60 mm	19959 Hz	19959 Hz
Stainless Steel 316	61.60 mm	61.08 mm	19780 Hz	19957 Hz
Al 6063-T52	61.60 mm	61.08 mm	19799 Hz	19976 Hz
Al 2024-T4	61.60 mm	61.08 mm	19805 Hz	19983 Hz

Amplifying Horn Original Dimensions

Amplifying Horn	Large Cylinder Length	Small Cylinder Length	Curve Axial Length	Horn Type
Al 6061 v1	40.005 mm	35.052 mm	59.944 mm	Exponential Horn
Ti 64 v1	40.000 mm	25.000 mm	60.000 mm	Exponential Horn
Ti 64 v2	65.532 mm	60.960 mm	8.890 mm	Exponential Horn

Amplifying Horn Optimized Dimensions

Amplifying Horn	Large Cylinder Length	Small Cylinder Length	Curve Axial Length	Horn Type
Al 6061 v1a	63.410 mm	65.240 mm	4.000 mm	Cosine Horn
Al 6061 v1b	55.730 mm	25.200 mm	61.280 mm	Conical Horn
Ti 64 v1a	61.990 mm	63.780 mm	4.000 mm	Cosine Horn
Ti 64 v1b	55.540 mm	22.920 mm	61.280 mm	Conical Horn
Ti 64 v2a	62.450 mm	63.000 mm	4.000 mm	Cosine Horn

Conclusion

- The designs that worked were the ones whose natural frequencies reached 20000 Hz
- The most optimal design would have to have a high gain and reach 20000 Hz
- The most optimal designs were the ones with a cosine curve and near-identical cylinder lengths